Unit 2 Reflection

What?

Unit 2 focused on **Exploratory Data Analysis (EDA)** and its role within the data science lifecycle. Through lectures, tutorials, and practical exercises, I developed a strong understanding of the purpose, techniques, and significance of EDA in preparing data for analysis and modelling. EDA is a critical stage because it enables analysts to identify patterns, detect anomalies, test assumptions, and generate hypotheses for further investigation. It also facilitates a deeper understanding of the dataset’s structure, variables, and relationships before applying machine learning models.

Another important aspect of EDA is its contribution to data preparation. Through this process, issues such as missing values, outliers, inconsistencies, and potential biases can be identified and addressed, helping to ensure that subsequent analyses and predictive models are based on high-quality data. As a result, EDA serves as a foundation for making informed modelling decisions and improving overall analytical outcomes.

So What?

A significant insight from this unit emerged when I compared the EDA process I had previously conducted in a module focused on statistical linear regression with the EDA undertaken in preparation for deep learning models. This comparison highlighted the differing requirements and assumptions associated with traditional statistical methods and modern machine learning approaches.

In classical linear regression, considerable emphasis is placed on validating assumptions such as normality, linearity, homoscedasticity, and independence of errors. If these assumptions are violated, analysts often need to transform the data or select a more appropriate modelling technique, such as logistic regression or other non-linear approaches. Consequently, EDA in traditional statistical modelling is heavily focused on verifying whether the data satisfies these underlying requirements.

By contrast, deep learning models are generally more flexible and impose fewer assumptions regarding the distribution or relationships within the data. However, this flexibility does not eliminate the need for thorough EDA. Issues such as class imbalance, missing values, noisy observations, and data quality concerns can still have a substantial impact on model performance. This comparison helped me appreciate the trade-off between the interpretability and statistical rigor of traditional methods and the adaptability and predictive power of deep learning approaches. It also reinforced the importance of tailoring EDA techniques to the specific modelling framework being used.

Now What?

One of the most valuable outcomes of this unit was gaining practical experience conducting EDA in **Python** using widely adopted libraries such as **NumPy**, **Pandas**, **Matplotlib**, and **Seaborn**. These tools enabled me to perform data manipulation, statistical analysis, and data visualization efficiently while exploring the characteristics of complex datasets.

Prior to this module, my experience with data analysis and visualization was primarily based in **R**. Learning Python has therefore significantly expanded my analytical capabilities and increased my flexibility in selecting the most appropriate tools for different tasks. It has also strengthened my confidence in working within environments commonly used in machine learning and artificial intelligence projects.

Moving forward, I intend to continue using **R** for conventional statistical analysis and data visualization because of its cohesive ecosystem and strong support for statistical methodologies. However, for advanced analytics, machine learning, and deep learning applications, I will increasingly rely on **Python** due to its extensive range of libraries, broader industry adoption, and seamless integration with modern machine learning frameworks. By developing proficiency in both languages, I will be better equipped to select the most effective tool for a given problem and adapt to a wide variety of analytical and professional contexts.